



Investigating the Reactor Antineutrino Anomaly (RAA)

Explained, to the best capabilities of **Sory Kaba**

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Elemental Particles

Fermions

Fermions have a particle spin equal to a half-integer value. The two basic constituents of matter are quarks and leptons.

Leptons: Electron Electron Neutrino
Muon, Muon Neutrino, Tau, Tau
Neutrino

Quarks: Quarks – up, down, charmed, strange, top, bottom (physicists think these names are funny!)

Hadrons: combinations of quarks - protons and neutrons

Bosons

Bosons have a particle spin equal to an integer

Photon, W, Z & Higgs Boson,
Gluon and Graviton

Photon: particle of light defined as a discrete bundle (or *quantum*) of electromagnetic (or light) energy. Photons are always in motion and, in a vacuum (a completely empty space), have a constant speed of light to all observers. Photons travel at the vacuum speed of light (more commonly just called the speed of light) of $c = 2.998 \times 10^8 \text{ m/s}$.



AntiMatter

AntiMatter

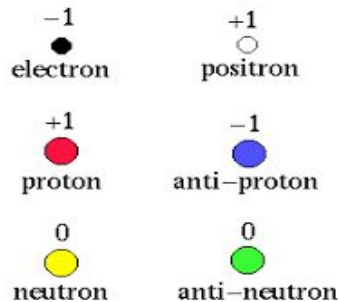
Every elementary particle has a corresponding antiparticle, which is antimatter.

Antimatter particles share the same mass as their matter counterparts, but qualities such as electric charge are opposite. The positively charged positron, for example, is the antiparticle to the negatively charged electron. Matter and antimatter particles are always produced as a pair and, if they come in contact, annihilate one another, leaving behind pure energy.

Antimatter

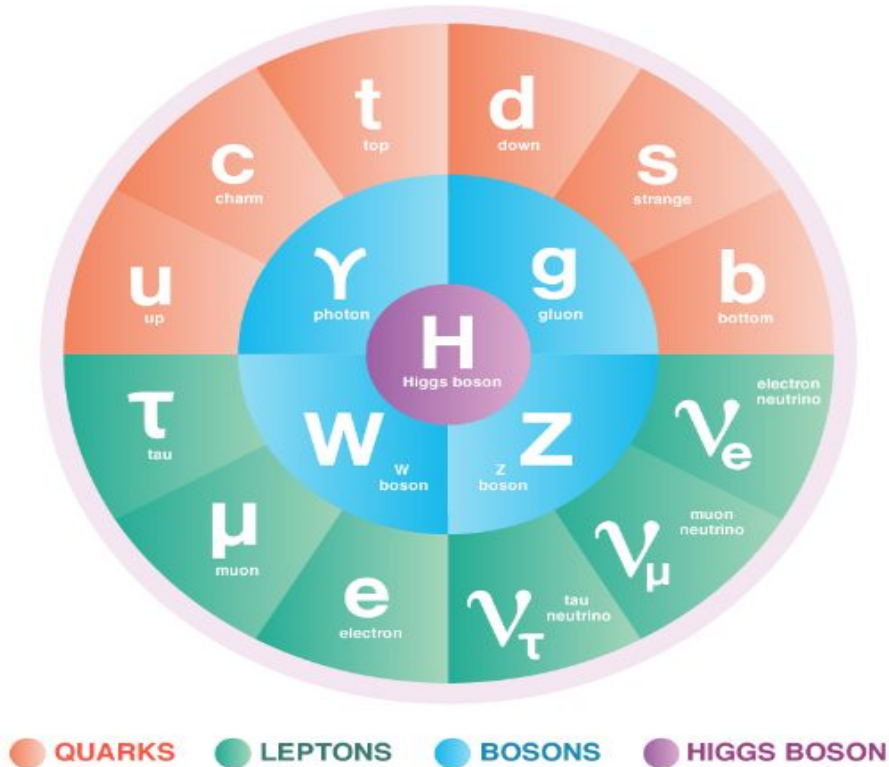
All elementary particles have antimatter counterparts. Antimatter counterparts have the same mass, but opposite electric charge.

NAME	SYMBOL	CHARGE	MASS (AMU)	MASS (G)
proton	p^+	1+	1.007	1.673×10^{-24}
neutron	n^0	0	1.009	1.675×10^{-24}
electron	e^-	1-	5.486×10^{-4}	9.109×10^{-28}



When matter and antimatter come in contact they are instantly converted into energy (annihilation).

Standard Model of Particle Physics





Neutrinos

Neutrino

The neutrino is an elementary particle that has **no electrical charge**, travels at (nearly) the speed of light, and passes through ordinary matter with virtually **no interaction**.

Neutrinos are produced through radioactive interactions, such as solar fusion, supernovae, radioactive decay, and when cosmic rays collide with the Earth's atmosphere.

There are actually three types (“flavors” - funny again) of neutrino:

electron neutrino, **muon neutrino**, and **tau neutrino**.

Antineutrino Anomaly

Neutrinos actually have some extremely bizarre behavior and are extensively studied for insight into physics “Beyond the Standard Model” (BSM physics).

“Neutrino oscillations” occur in which one flavor of neutrino changes into a different flavor. This is BSM physics and is of tremendous interest to particle physics.

To study neutrino oscillations one can use nuclear reactors, because they are tremendous sources of neutrinos.

The energy spectrum of neutrinos from a reactor is essential for these studies. It should be derivable from known nuclear data.
But the answer comes out wrong!

This is the Reactor Antineutrino Anomaly, which we are attempting to explain.



Radiation

Radioactive Decay

Radioactive decay occurs when an unstable atomic nucleus loses energy by emitting energy in the form of emitted particles or electromagnetic waves, called radiation. Isotopes are atoms of the same element (thereby having the same number of protons) which differ in the number of neutrons in their nucleus. Some isotopes of a given element are more unstable than others, causing a nuclear reaction which releases energy to achieve a more stable nuclear configuration. Such isotopes are radioactive, and are referred to as “radioisotopes.”

Alpha Ray and Decay

An alpha particle (α /alpha) is made up of two protons and two neutrons bound together. This type of radiation has a positive charge (due to the presence of two protons). An alpha particle is sometimes represented using the chemical symbol He^{2+} , because it has the same structure as a helium atom missing its two electrons—hence the overall charge of +2. Their massive size (compared to beta particles, for instance) means alpha particles have very low penetration power. Penetration power describes how easily the particles can pass through another material. Since alpha particles have a low penetration power, the outside layer of the human skin, for example, can block these particles.

Alpha decay occurs because the nucleus of a radioisotope has too many protons. A nucleus with too many protons causes repulsion between these like charges. To reduce this repulsion, the nucleus emits an α particle. Examples of this can be seen in the decay of americium (Am) to neptunium (Np).

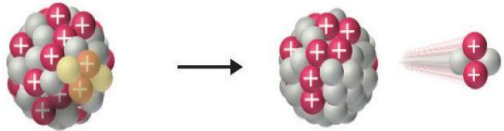
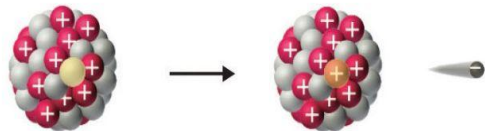
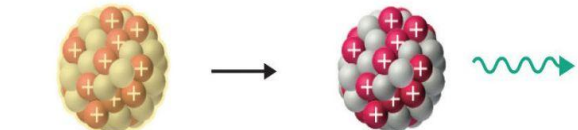
Beta Rays and Decay

In radioactive nuclei with too many neutrons, a neutron can be converted into an electron, called beta particle. Beta particles (β) have a higher penetration power than alpha particles (they are able to pass through thicker materials such as paper).

During beta decay, the number of neutrons in the atom decreases by one, and the number of protons increases by one. Effectively, a neutron was converted into a proton in the decaying nucleus, in the process releasing a beta particle. Since the number of protons before and after the decay is different, the atom has changed into a different element.

Gamma Ray and Decay

Gamma decay is one type of radioactive decay that a nucleus can undergo. What separates this type of decay process from alpha or beta decay is that no particles are ejected from the nucleus when it undergoes this type of decay. Instead, a high energy form of electromagnetic radiation - a gamma ray photon - is released. Gamma rays are simply photons that have extremely high energies which are highly ionizing.[1] As well, gamma radiation is unique in the sense that undergoing gamma decay does not change the structure or composition of the atom. Instead, it only changes the energy of the atom since the gamma ray carries no charge nor does it have an associated mass.

Decay Type	Radiation Emitted	Generic Equation	Model
Alpha decay	${}^4_2\alpha$	${}_Z^AX \longrightarrow {}_{Z-2}^{A-4}X' + {}^4_2\alpha$	 Parent Daughter Alpha Particle
Beta decay	${}^0_{-1}\beta$	${}_Z^AX \longrightarrow {}_{Z+1}^AX' + {}^0_{-1}\beta$	 Parent Daughter Beta Particle
Gamma emission	${}^0_0\gamma$	${}_Z^AX^* \xrightarrow{\text{Relaxation}} {}_Z^AX' + {}^0_0\gamma$	 Parent (excited nuclear state) Daughter Gamma ray

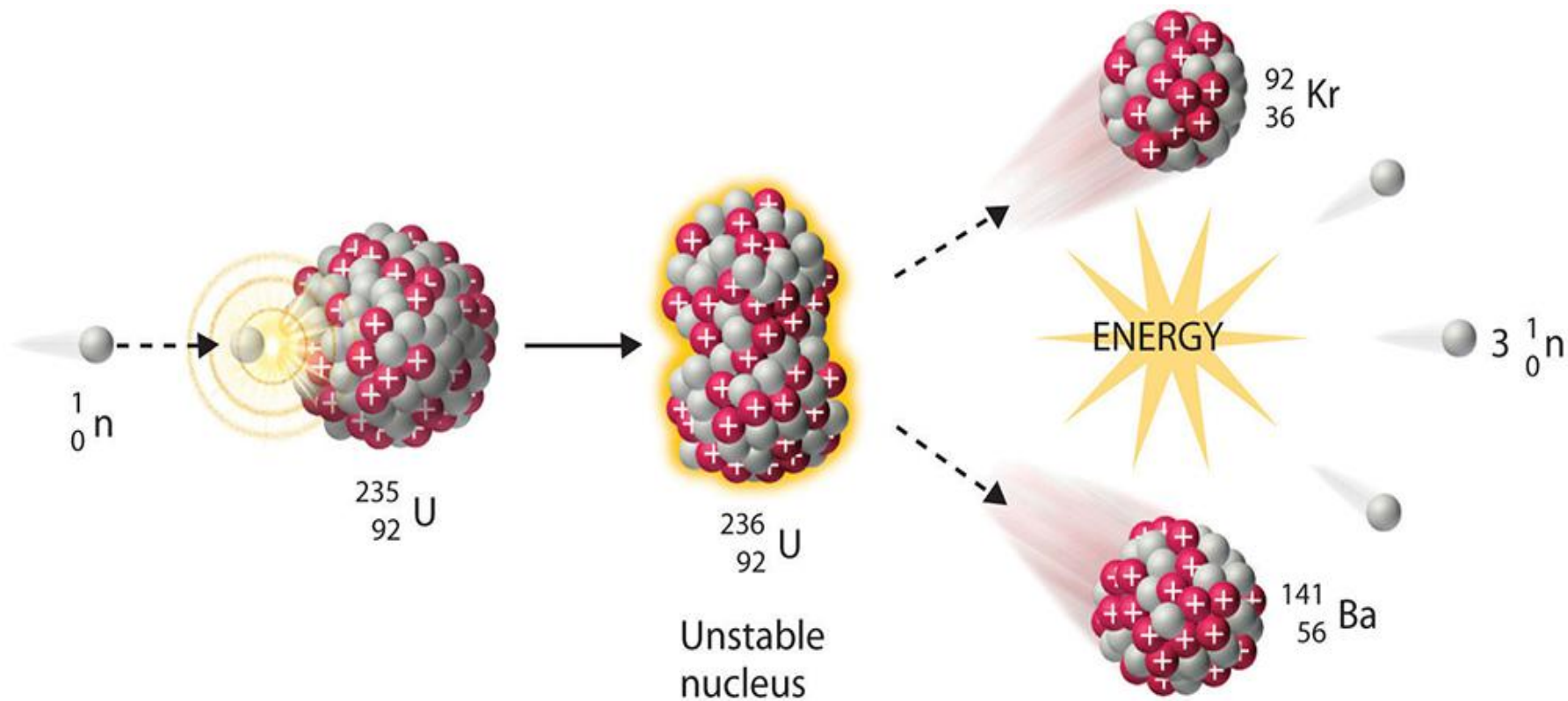


Fission and Fission Yield

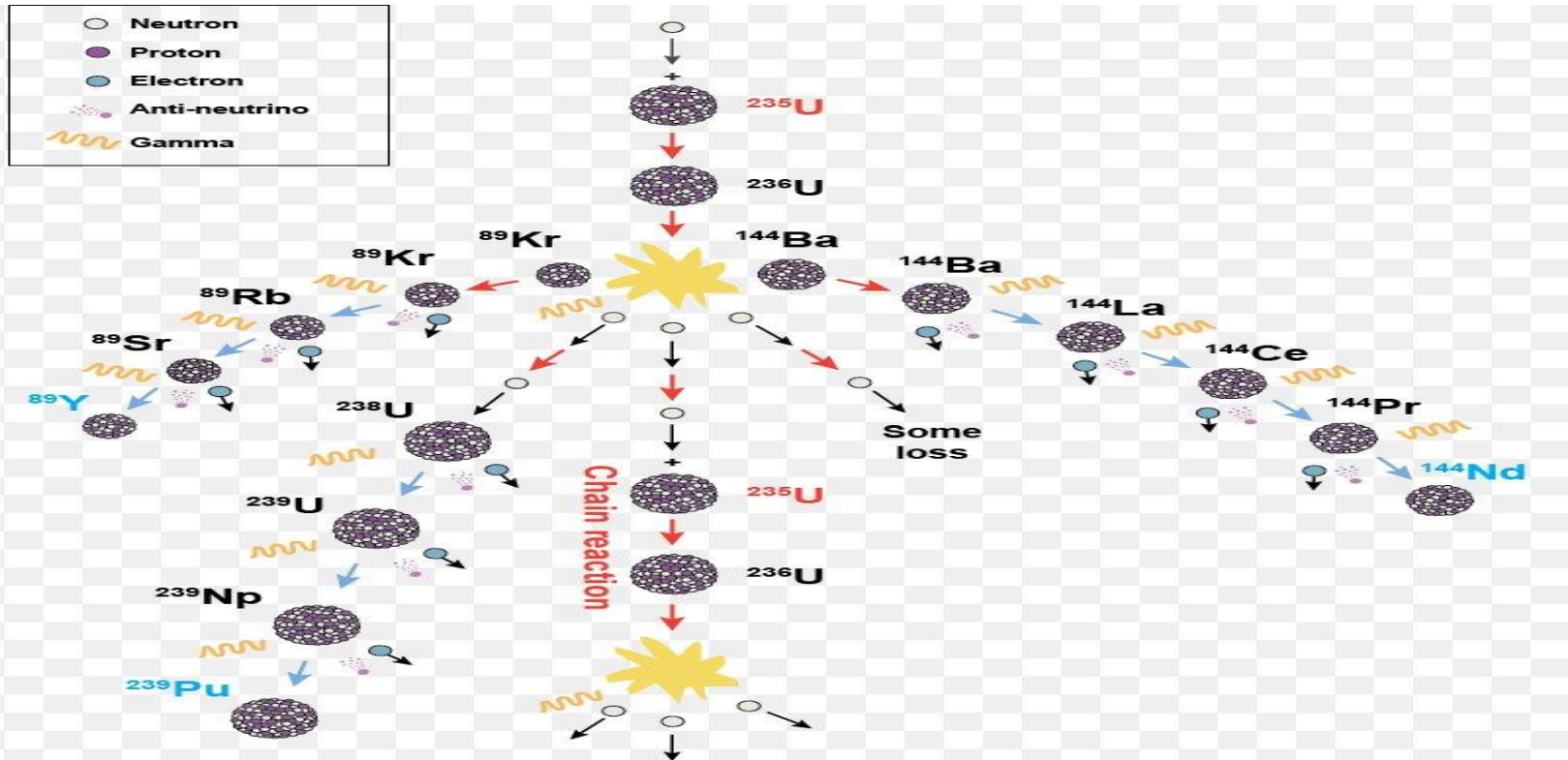


Fission

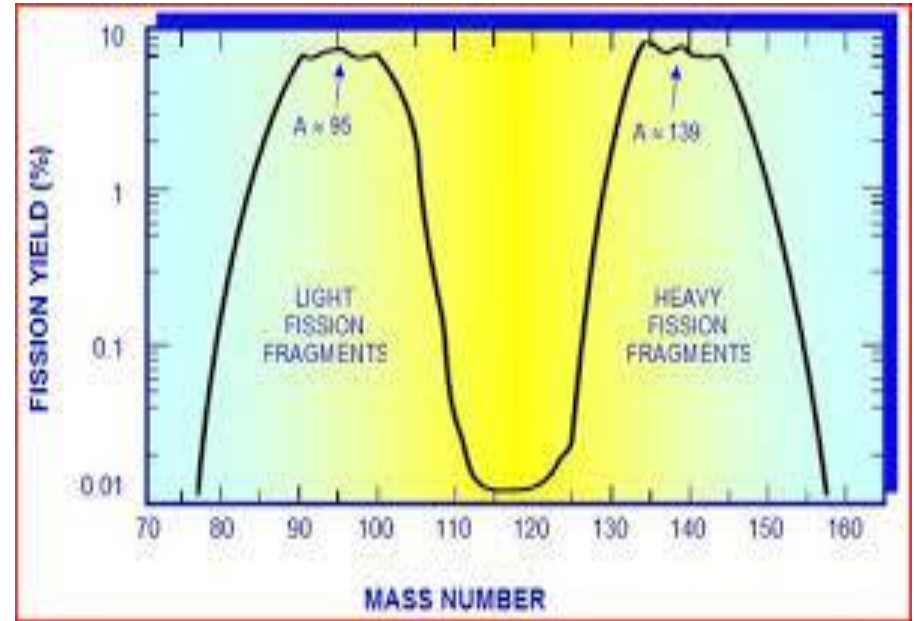
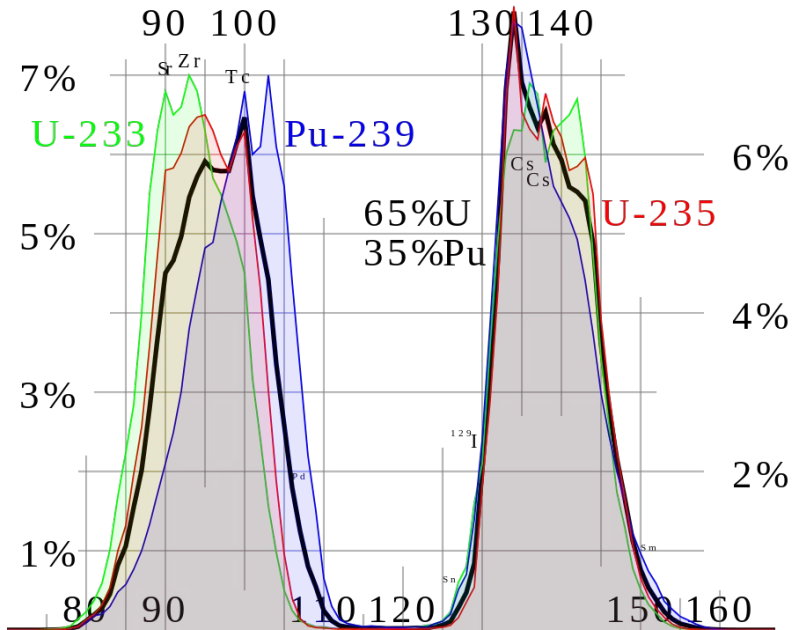
- * Nuclear fission is a reaction in which the nucleus of an atom splits into two or more smaller nuclei. The fission process often produces several gamma rays and neutrons, and releases a very large amount of energy even by the energetic standards of radioactive decay. It also produces the two fragments of the original nucleus, which are called “fission products”.
- * Fission products beta decay, which is where the reactor anti-neutrinos come from.
- * Cumulative fission yields: Give the amounts of each nuclide produced either directly in the fission or by decay of other fission products. **Errors in tabulated cumulative fission yields may be the reason the calculated antineutrino spectrum comes out wrong. We want to check these.**



Fission Chain



Fission Yield





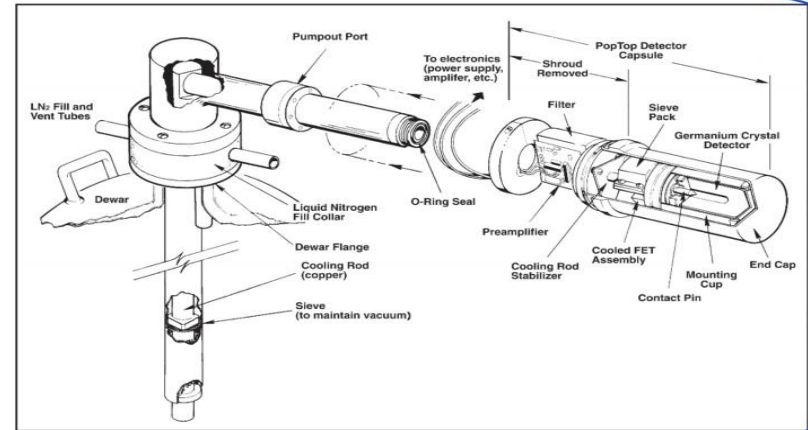
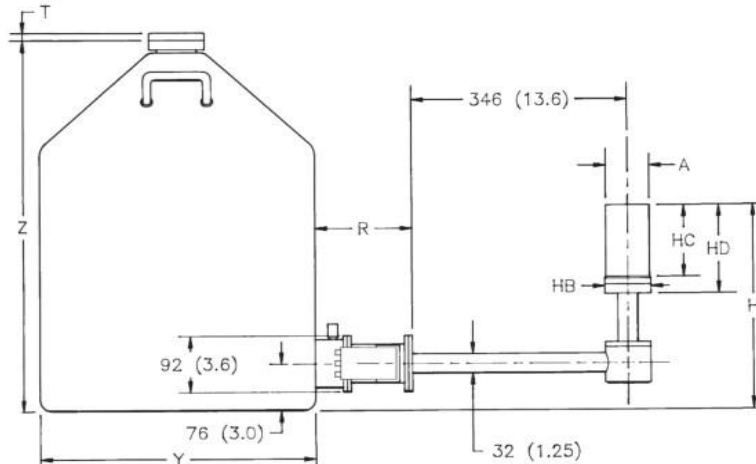
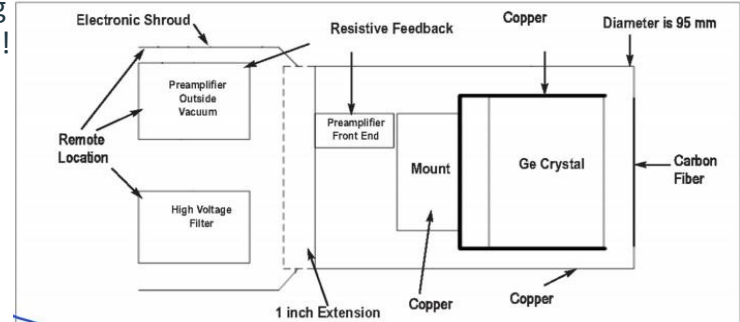
Introducing the GMX

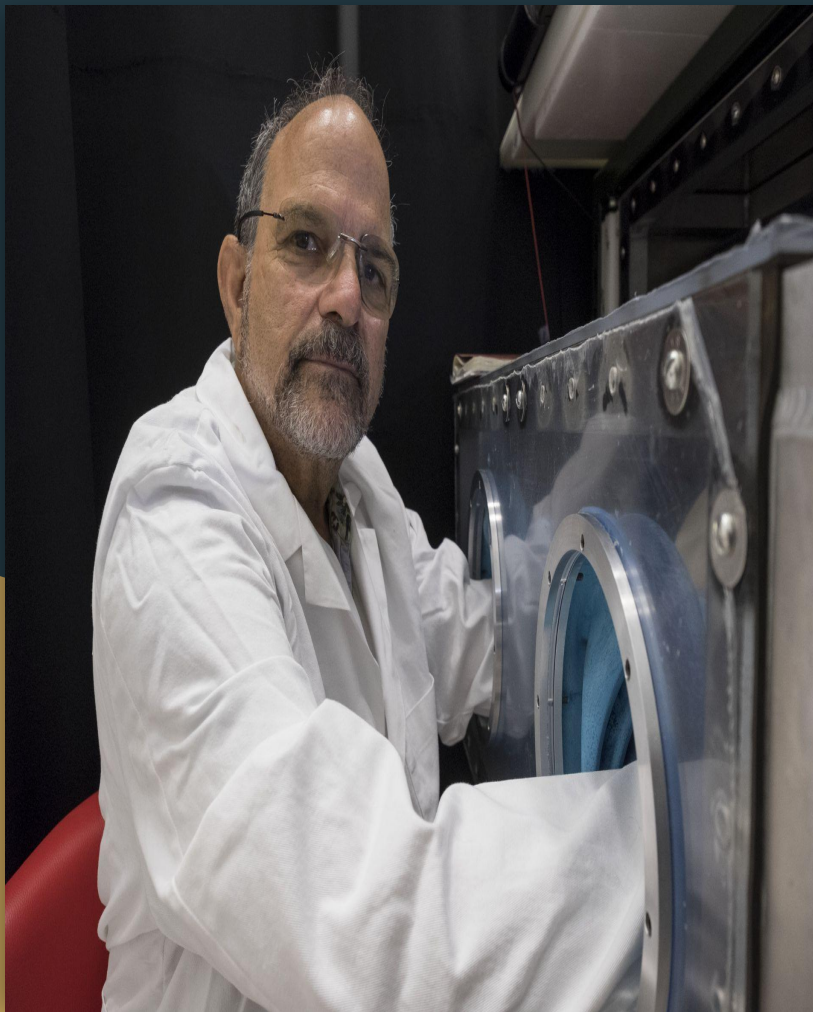
High Purity Germanium Detector Components

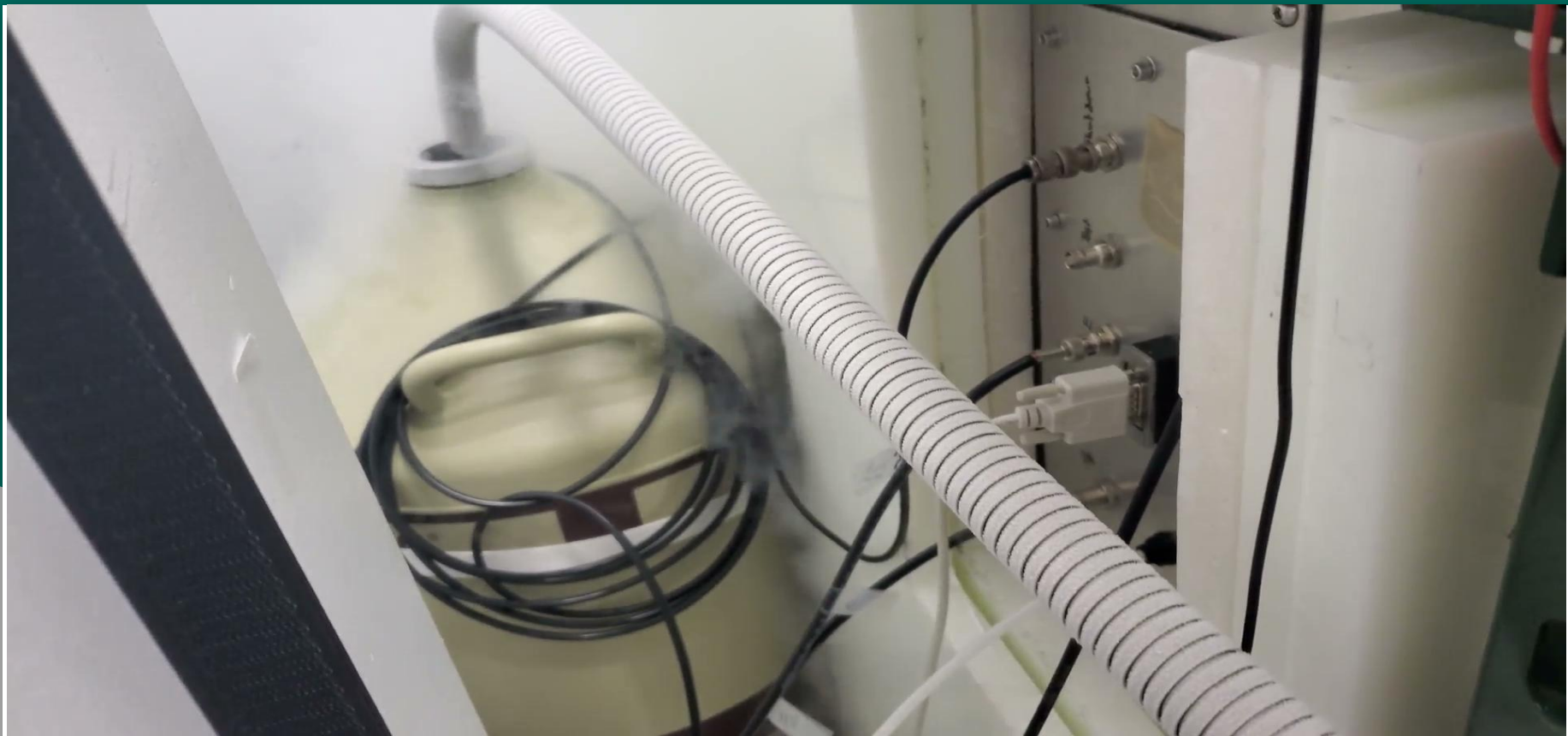
We use the GMX to measure gamma rays from the chains of decaying fission products. This allows us to check the cumulative fission yields! GMX dewar and cryostat configuration

A diagram of the GMX endcap

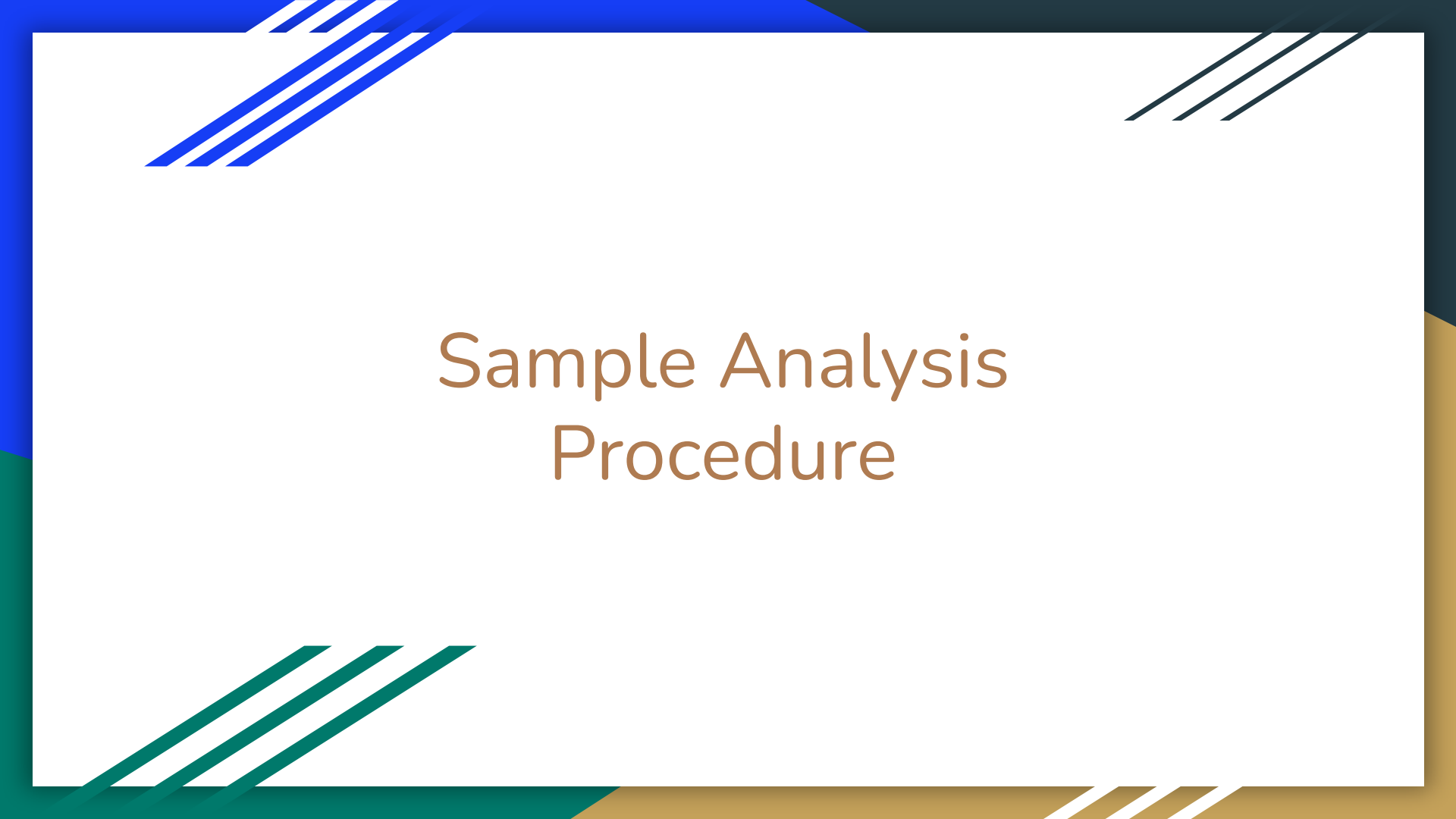
Material compositions within the GMX endcap







The Refilling process

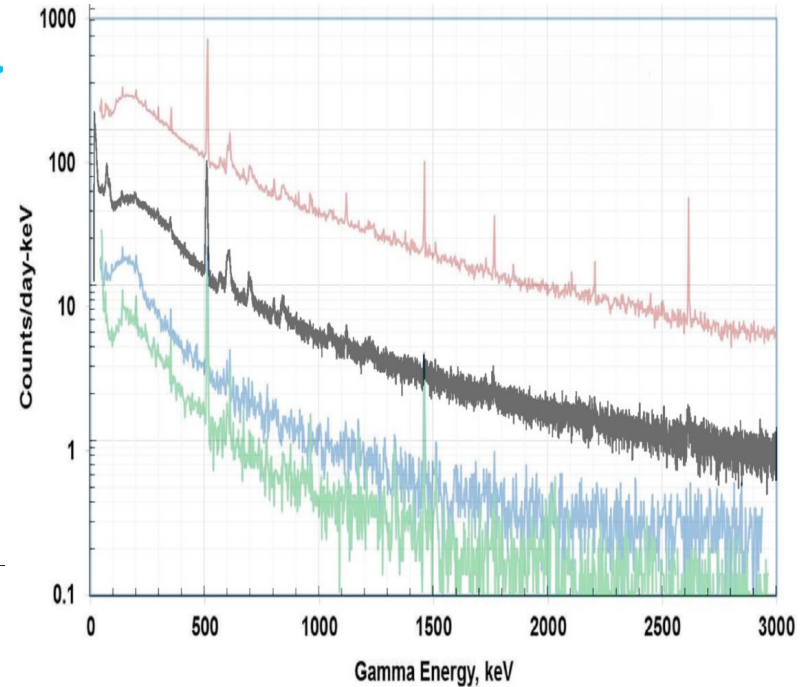
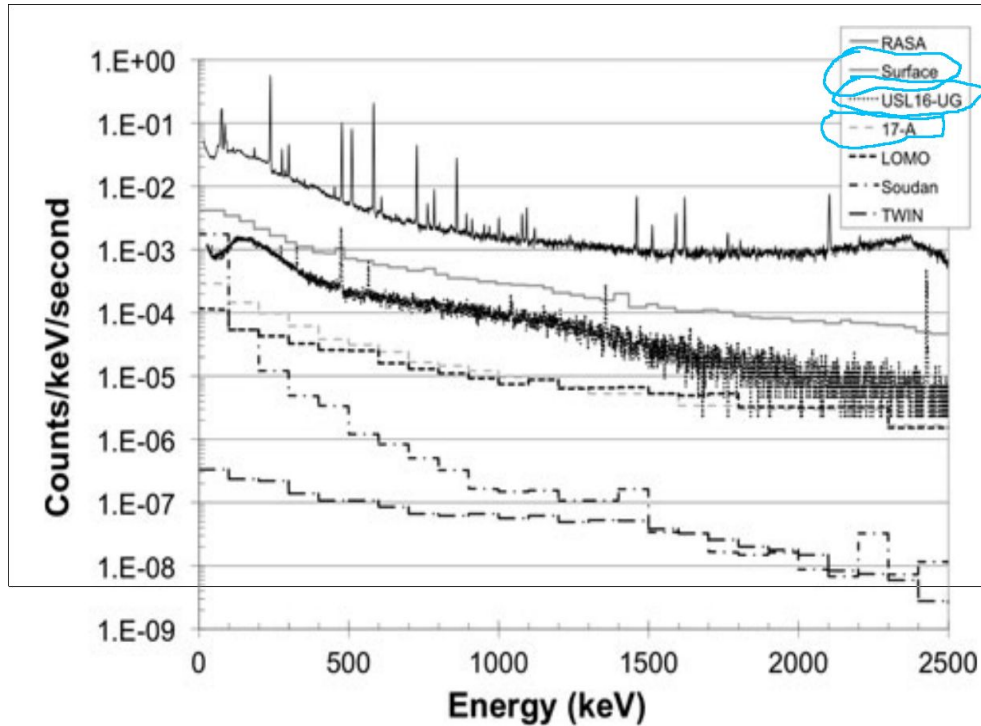


Sample Analysis Procedure

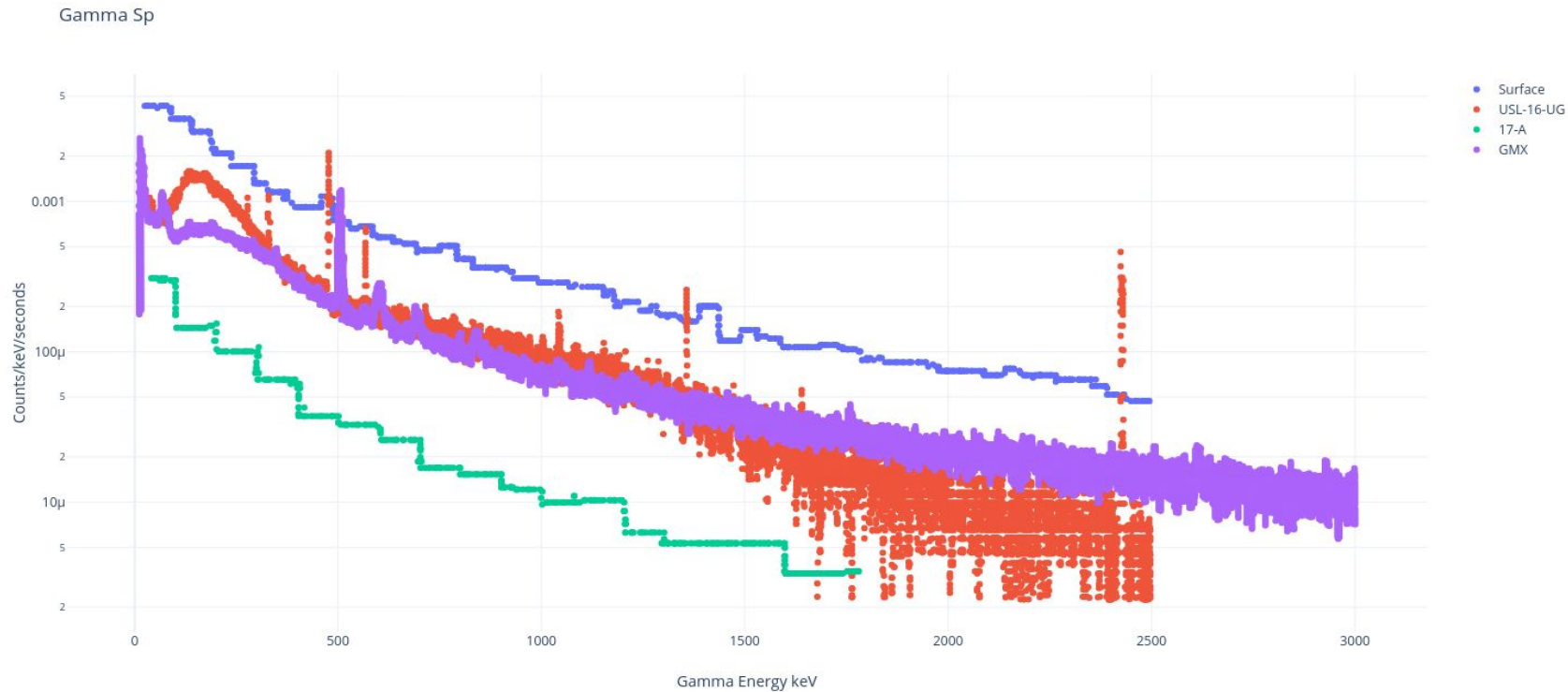
Sample Analysis Procedure

- 1) 1)A program called RadICalc uses tabulated nuclear data (which we want to check) and calculates the expected gamma ray spectrum emitted from reactor fuel samples (we got them from ORNL)
- 2) 2)Gather the observed gamma ray spectra with a detector like GMX
- 3) 3)Use data and simulations on the detector to convert observed gammas into emitted gammas
- 4) 4)Compare with RadICalc predictions
- 5) 5)Publish

Comparison Charts



GMX EFFICIENCIES



Prominent Beta Decay Chains

TABLE I. Most prominent beta decay branches with $E_\beta = 5\text{--}7$ MeV. The table presents the decay parent, end-point energy, half-life, and decay ft value. The decay type describes the parent and daughter states. The moderate ft values and lack of significant change of J^π suggest that all but possibly ^{140}Cs decay with allowed spectral shapes. The rate each branch contributes to the total between 5–7 MeV is N , accounting for the inverse beta decay cross section. The 1σ uncertainty due to the fission yield and branching fraction is σ_N .

Isotope	Q(MeV)	$t(s)$	$\text{Log}(ft)$	Decay type	N (%)	σ_N (%)
^{96}Y	7.103	5.34	5.59	$0^- \rightarrow 0^+$	13.6	0.8
^{92}Rb	8.095	4.48	5.75	$0^- \rightarrow 0^+$	7.4	2.9
^{142}Cs	7.308	1.68	5.59	$0^- \rightarrow 0^+$	5.0	0.7
^{97}Y	6.689	3.75	5.70	$1/2^- \rightarrow 1/2^+$	3.8	1.1
^{93}Rb	7.466	5.84	6.14	$5/2^- \rightarrow 5/2^+$	3.7	0.5
^{100}Nb	6.381	1.5	5.1	$1^+ \rightarrow 0^-$	3.0	0.8
^{140}Cs	6.220	63.7	7.05	$1^- \rightarrow 0^+$	2.7	0.2
^{95}Sr	6.090	23.9	6.16	$1/2^+ \rightarrow 1/2^-$	2.6	0.3



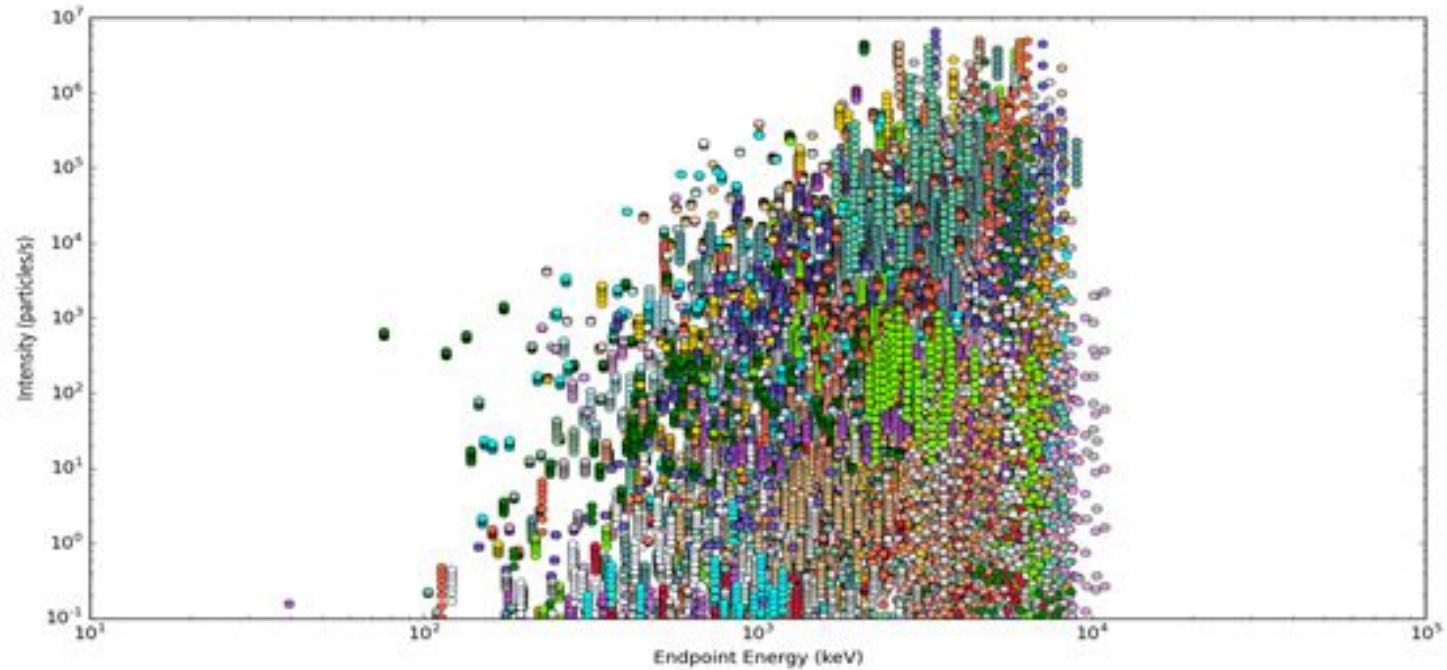
Analysis of ^{235}U



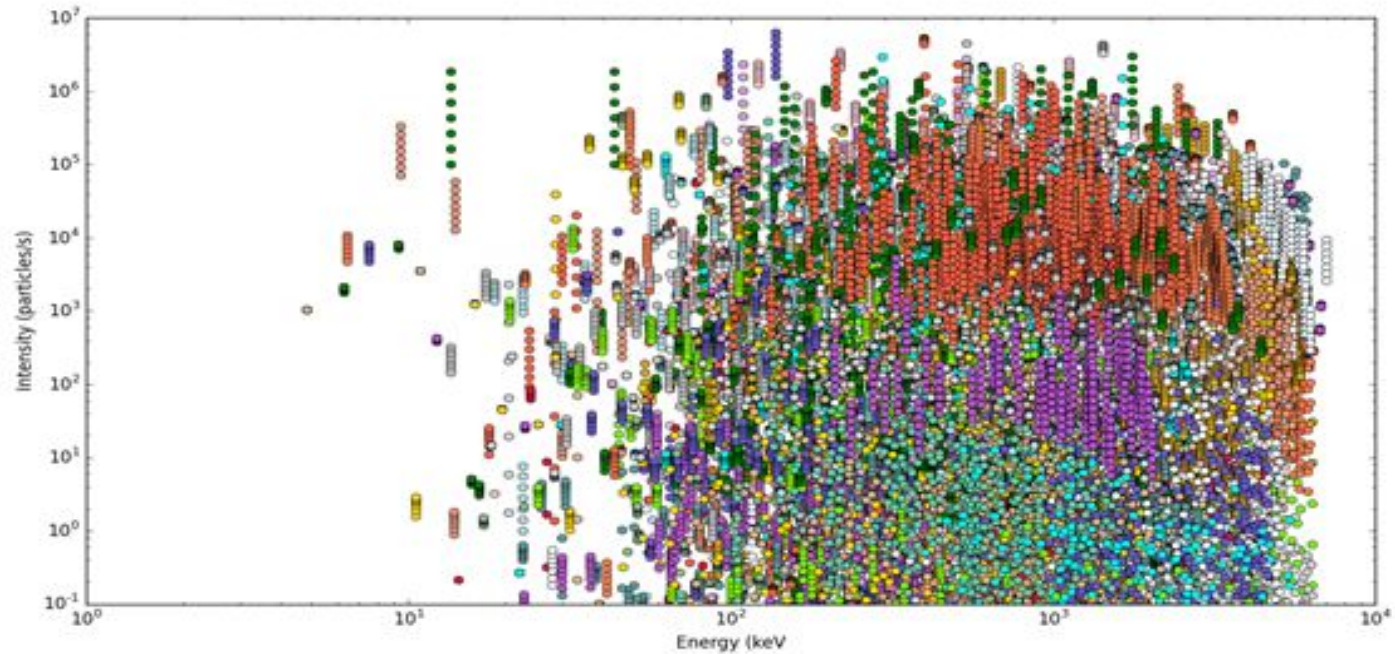
Cumulative Fission Yield Data Set

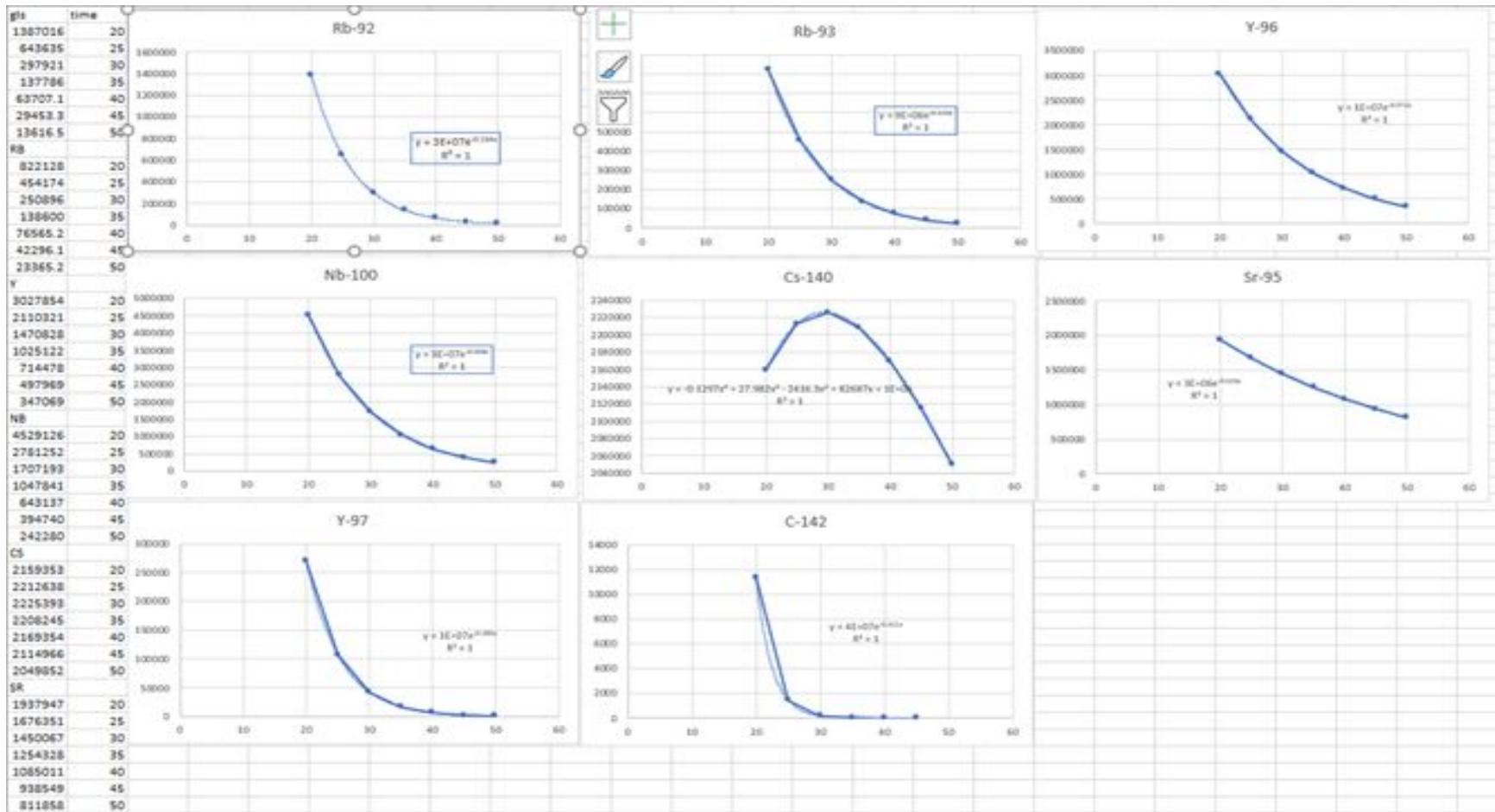
Isotope	Atomic n	Mass (amu)	Energy (keV)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)	Intensity (Uncertainty)
Cu-68	29	68	1077.7	0.303909	1.550232	0.338313	1.723081	0.361616	1.840271	0.375829	1.911765	0.382657	1.946034	0.383544	1.950263	0.379702	1.930532
Ni-69	28	69	1213	1.036867	3.960927	0.765054	2.918505	0.564497	2.149762	0.416515	1.583023	0.307327	1.165339	0.226762	0.857602	0.167316	0.630942
Ni-69	28	69	470.5	0.335149	1.279396	0.24729	0.942688	0.182464	0.69438	0.134631	0.511321						
Ni-69	28	69	1483.2	0.900712	3.438593	0.664593	2.533634	0.490371	1.866265	0.361821	1.374263	0.266971	1.011659	0.196985	0.744504	0.145346	0.547734
Ni-69	28	69	1710.9	0.408463	1.559939	0.301385	1.1494	0.222378	0.846644	0.164082	0.623444	0.121068	0.458947				
Ni-69	28	69	679.7	1.04734	3.998032	0.772782	2.945842	0.570199	2.169895	0.420722	1.597847	0.310431	1.176249	0.229052	0.865629	0.169007	0.636846
Ni-69	28	69	574.6	0.335149	1.280378	0.24729	0.943413	0.182464	0.694915	0.134631	0.511716						
Ni-69	28	69	249.1	0.115207	0.440632												
Ni-69	28	69	205.4	0.701718	2.678675	0.517764	1.973709	0.382033	1.453826	0.281884	1.070555	0.207989	0.788085	0.153465	0.57997	0.113234	0.426686
Ni-69	28	69	1871.1	1.07876	4.118629	0.795965	3.034701	0.587305	2.235349	0.433344	1.646046	0.319744	1.211731	0.235924	0.891742	0.174077	0.656057
Ni-69	28	69	183.2	0.104734	0.400756												
Cu-69	29	69	1007.5	0.102467	0.466361	0.111086	0.497689	0.116733	0.517726	0.120203	0.52949	0.122079	0.535207	0.122794	0.536519	0.122665	0.534639
Cu-70	29	70	884.88	3.682031	12.0211	3.622137	11.86723	3.479081	11.4645	3.296844	10.93955	3.099615	10.3627	2.900513	9.772677	2.706465	9.190446
Cu-70	29	70	1954.2	0.103097	0.336721	0.10142	0.33241										
Cu-70	29	70	2061.6	0.18042	0.589106	0.177485	0.581565	0.170475	0.561829	0.161545	0.536102	0.151881	0.507833	0.142125	0.478918	0.132617	0.450385
Cu-70	29	70	901.7	3.670985	11.98504	3.61127	11.83163	3.468644	11.43011	3.286953	10.90673	3.090316	10.33161	2.891811	9.743359	2.698346	9.162875
Cu-70	29	70	1108.4	0.342429	1.117997	0.336859	1.103686	0.323555	1.066231	0.306606	1.017408	0.288264	0.96376	0.269748	0.908886	0.251701	0.854737
Cu-70	29	70	1251.7	2.698929	8.811472	2.655026	8.698681	2.550167	8.403483	2.416586	8.018689	2.272018	7.595862	2.126076	7.163374	1.983839	6.736599
Cu-70	29	70	1690.3	0.231968	0.757384	0.228195	0.747689	0.219182	0.722315	0.207701	0.68924	0.195276	0.652896	0.182732	0.615722	0.170507	0.579038
Zn-71	30	71	910.3	0.133959	0.691521	0.156047	0.800686	0.173562	0.886954	0.187238	0.954156	0.197712	1.005505	0.205528	1.043688	0.211139	1.070948
Zn-71	30	71	511.6	0.546773	2.82436	0.636926	3.270231	0.708418	3.622585	0.764236	3.897063	0.80699	4.106795	0.838889	4.262748	0.86179	4.37409
Zn-71	30	71	390											0.100667	0.511222	0.103415	0.524575
Cu-72	29	72	2859.2	0.673782	3.280133	0.399653	1.945595	0.236972	1.153485	0.140502	0.683793						
Cu-72	29	72	858.3	0.464677	2.261111	0.275623	1.341167	0.163429	0.795138								
Cu-72	29	72	846.75	2.648661	12.88906	1.57105	7.645081	0.931543	4.532539	0.552317	2.686916	0.327468	1.592753	0.194154	0.944125	0.115114	0.559627
Cu-72	29	72	1251.5	1.765774	8.592706	1.047366	5.096722	0.621029	3.021693	0.368211	1.791278	0.218312	1.061835	0.129436	0.629417		
Cu-72	29	72	1540	1.579903	7.689298	0.937117	4.56087	0.555657	2.704003	0.329452	1.602949	0.195332	0.950198	0.115811	0.563242		
Cu-72	29	72	2040.5	0.30204	1.470757	0.179155	0.872372	0.106229	0.517203								
Cu-72	29	72	2540	0.255573	1.245084	0.151592	0.738515										
Cu-72	29	72	652.68	23.23387	113.0556	13.78114	67.0584	8.171431	39.75691	4.844884	23.56813	2.875222	13.97074	1.703109	8.281341	1.009768	4.908738
Cu-72	29	72	3941.6	0.325274	1.588601	0.192936	0.942271	0.1144	0.558645								
Cu-72	29	72	3008.9	0.836419	4.070495	0.496121	2.414394	0.294172	1.431422	0.174416	0.848555	0.103411	0.503007				
Cu-72	29	72	988.24	0.441444	2.148175	0.261842	1.27418	0.155257	0.755423								
Cu-72	29	72	1993.1	1.626371	7.915139	0.96468	4.694826	0.572	2.783422	0.339142	1.650029	0.201077	0.978106	0.119218	0.579785		
Cu-72	29	72	2769.8	0.255573	1.245084	0.151592	0.738515										

Beta Spectrum of 235-U



Gamma Spectrum of ^{235}U





Total g/s emitted at 30 secs

Isotope	Energy ke	20sec	25sec	30sec	35sec	40sec	45sec	50sec	Total
Rb-92	814.98	1387016	643634.7	297921	137785.5	63707.11	29453.27	13616.52	8.86E+06
Rb-93	432.61	822128.3	454174.4	250895.7	138599.7	76565.16	42296.07	23365.17	6.80E+06
Y-96	1750.6	3027854	2110321	1470828	1025122	714478.4	497969.4	347069.3	2.91E+07
Nb-100	535.666	4529126	2781252	1707193	1047841	643137	394739.6	242280.1	4.08E+07
Cs-140	602.25	2159353	2212638	2225393	2208245	2169354	2114966	2049852	5.72E+07
Sr-95	685.6	1937947	1676351	1450067	1254328	1085011	938549.3	811858	3.37E+07
Y-97	3287.6	270146.2	107207.6	42545.35	16884.14	6700.473	2659.085	1055.258	1E+06
Cs-142	359.598	11373.89	1468.197	188.4331	24.11976	3.083546	0.393982		<u>2.62E+04</u>

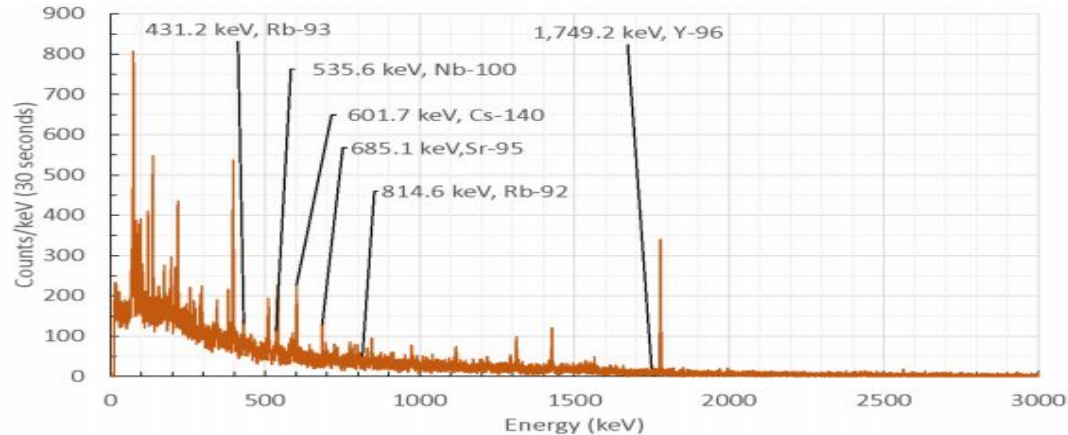
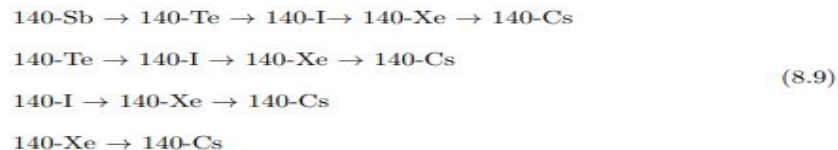


Figure 8.4: Ge gamma ray spectrum from ORNL irradiation of ^{235}U . The strongest gamma energies from the six beta decay chains of interest are identified in the spectrum.

where number of nuclides fission produced (N_{fi}) and fission yield ($Y_{fission}$) are used. Fission yield data values are based on ENDF VII [68], [69]. Decay after the irradiation involves complicated decay schemes. For example, Equation 8.9 shows the linear decay chains of ^{140}Cs from its predecessors that are fission produced [70].



Manual Calculations using Batemans Equations

Decay Equations

$$\frac{dN_p}{dt} = -\lambda_p N_p$$

$$\frac{dN_d}{dt} = -\lambda_d N_d + \lambda_p N_p$$

$$N_d(t) = N_p(0) \frac{\lambda_p}{\lambda_d - \lambda_p} \left(e^{-\lambda_p t} - e^{-\lambda_d t} \right) + N_d(0) e^{-\lambda_d t}$$

- Where
 - N_p is the # of atoms of parent
 - N_d is the # atoms of daughter
 - λ_p or λ_d is the decay constant ($\ln 2 / T_{1/2}$) of parent or daughter

$$A_p = \lambda_p N_p \quad \text{and} \quad A_d = \lambda_d N_d$$

$$A_d(t) = A_p(0) \frac{\lambda_d}{\lambda_d - \lambda_p} \left(e^{-\lambda_p t} - e^{-\lambda_d t} \right) + A_d(0) e^{-\lambda_d t}$$

- The second term in the activity equation is just the residual daughter product activity remaining from any that was present at $t=0$.
- This equation is simpler if we set this value to zero and just examine the first term. The equation for activity of the daughter is known as the Bateman equation

Overall process takes 2 steps:

Step 1) determine number of 140-Cs nuclide produced via decay chain at $t=20$ sec.

Step 2) determine gamma ray output at 602.25 keV based on 1).

The result of 2) should come close to but less than 2159353 g/s which is a RadlCalc output of 140-Cs at 602.25 keV at $t=20$ sec.

Details:

Step 1)

Each of four decay chains listed in Eq. 8.9 needs to be solved separately using decay equation which is an ordinary differential equation (ODE).

Here is an example for the first decay chain but the process needs to be applied to the rest of other decay chains as well.

The first decay chain consists of 5 coupled ODE that need to be solved simultaneously.

(Second decay chain as 4 coupled ODE, etc).

This ODE is an initial value problem: you will need to determine how many of $1E10$ fission products nuclides are 140-Sb & thus initially present for decaying.

There are many software available for solving this type of ODE.

You will also need to consider how many 140-Cs are present among $1E10$ fission nuclide and decay them for 20 sec as well.

So all together you will need to apply above process 5 times for each decay chain listed in Eq. 8.9 as well as 140-Cs itself.

As a reference, below lists the number of 140-Cs produced at $t=20$ after solving ODE.

The isotope listed is starting isotope from which 140-Cs is decayed from:

140-Sb 7.87

140-Te 925200.10

140-I 7484765.00

140-Xe 197770400.00

140-Cs 166454000.00

Adding them up, there are total of 372634372.97 140-Cs present at $t = 20$ sec.

Step 2)

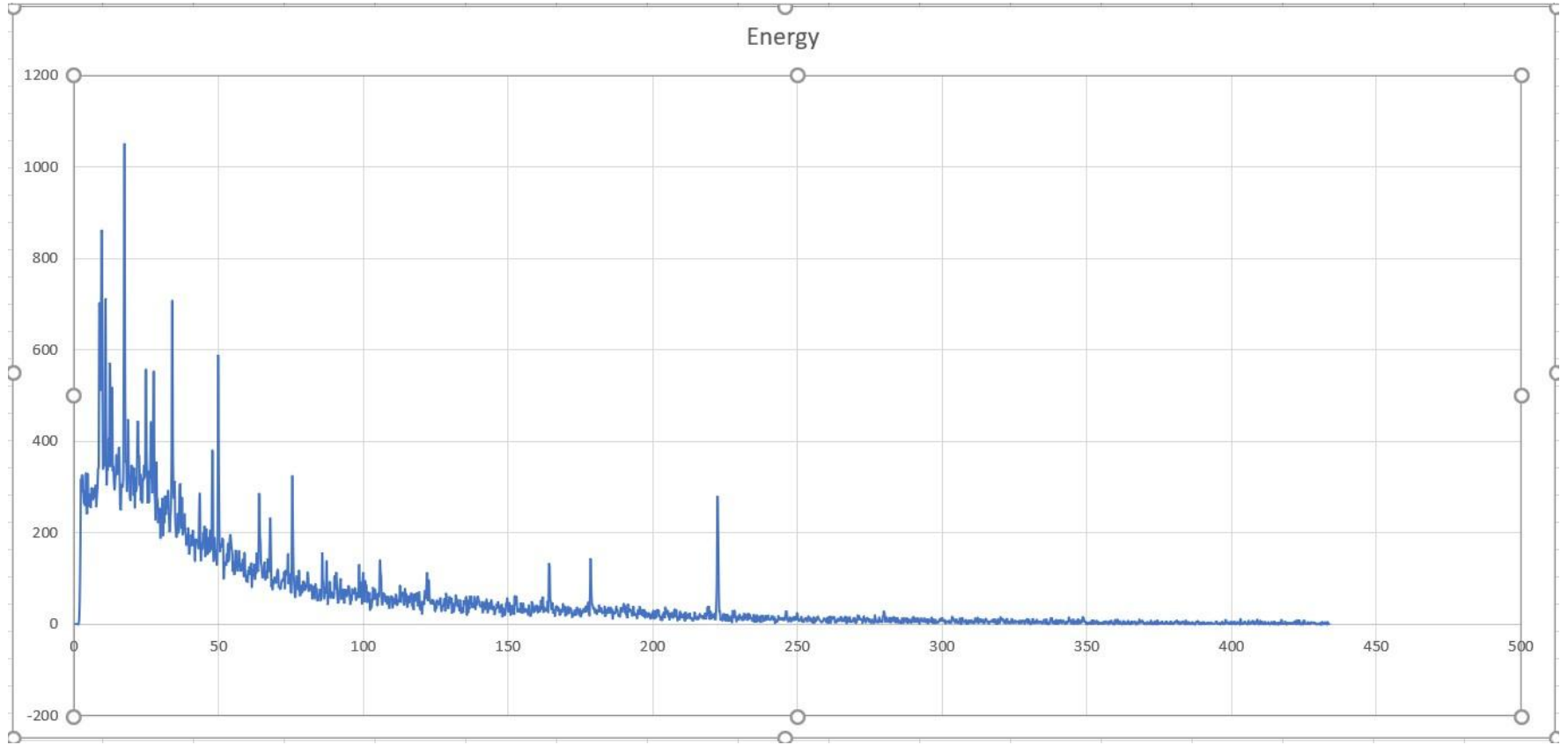
Based on 372634372.97 140-Cs nuclides, 1974962.18 g/s at 602.25 keV are produced.

You will need to study and understand how this conversion process comes about.

This is a relative error of 8.5% with respect to the RadlCalc output of 2159352.72 g/s.

This error is an upper bound.

Energy



Modifications of the GMX

Problems with the Gmx

- the glove department was too short not allowing someone to properly work with materials
- Needed to recalibrate parts of the detector for better reading
- Needed to build new shielding because old one became obsolete

References

Kim, Samuel. "Gamma Background Reduction in A Low Counting Facility At Shallow Depth, And Analysis of Gamma Emissions From Eight Beta Decay Contributors to the Reactor Antineutrino Spectrum Anomaly." *Temple University*, 5AD, pp. 1–125.

Jones, Andrew Zimmerman. "What Is a Photon in Physics?" ThoughtCo, Aug. 26, 2020, [thoughtco.com/what-is-a-photon-definition-and-properties-2699039](https://www.thoughtco.com/what-is-a-photon-definition-and-properties-2699039).

Jones, Andrew Zimmerman. "Physics: Fermion Definition." ThoughtCo, Feb. 16, 2021, [thoughtco.com/fermion-definition-in-physics-2699188](https://www.thoughtco.com/fermion-definition-in-physics-2699188).

Helmenstine, Anne Marie, Ph.D. "What Is Antimatter?" ThoughtCo, Feb. 16, 2021, [thoughtco.com/overview-of-antimatter-608646](https://www.thoughtco.com/overview-of-antimatter-608646).

Helmenstine, Anne Marie, Ph.D. "Subatomic Particles You Should Know." ThoughtCo, Feb. 16, 2021, [thoughtco.com/elementary-and-subatomic-particles-4118943](https://www.thoughtco.com/elementary-and-subatomic-particles-4118943).

Jones, Andrew Zimmerman. "Particle Physics Fundamentals." ThoughtCo, Feb. 16, 2021, [thoughtco.com/particle-physics-fundamentals-2698865](https://www.thoughtco.com/particle-physics-fundamentals-2698865).

Jones, Andrew Zimmerman. "Neutrinos in Particle Physics." ThoughtCo, Aug. 28, 2020, [thoughtco.com/neutrino-2698990](https://www.thoughtco.com/neutrino-2698990).

Wave-Particle Duality, Uncertainty Principle, abyss.uoregon.edu/~js/ast123/lectures/lec06.html.

References-Continued